

Imp

Power of a single-tone Amplitude-Modulated (AM) signal

Consider a carrier signal $A \cos \omega_c t$ is amplitude-modulated by a single-tone modulating signal $x(t) = V_m \cos \omega_m t$

• Unmodulated carrier power

$P_c = \text{mean square (ms value)}$

$$P_c = \overline{(A \cos \omega_c t)^2} = \frac{A^2}{2} \quad \text{same as } \frac{1}{2\pi} \int_0^{2\pi} V_m^2 \cos^2 \omega_m t$$

The sideband power $P_s = \frac{1}{2} \overline{x^2(t)} = \frac{1}{2} \overline{(V_m \cos \omega_m t)^2}$

$$= \frac{1}{2} \frac{V_m^2}{2} = \frac{1}{4} V_m^2$$

$$P_t = P_c + P_s$$

$$= \frac{A^2}{2} + \frac{1}{4} V_m^2$$

$$= \frac{A^2}{2} \left[1 + \frac{1}{2} \left(\frac{V_m}{A} \right)^2 \right]$$

$m_a = \text{modulation index, for AM}$

$$\frac{V_m}{A} = \frac{\text{max. baseband amplitude}}{\text{max. carrier amplitude}} = m_a = \text{modulation index for AM}$$

$$P_t = \frac{A^2}{2} \left[1 + \frac{1}{2} m_a^2 \right]$$

where But $\frac{A^2}{2} = P_c = \text{carrier power}$

$$P_t = P_c \left(1 + \frac{m_a^2}{2} \right)$$

Ex A 400 watts carrier is modulated to a depth of 75%.
Find the total power in the amplitude modulated wave.
Assume the modulating signal to be a sinusoidal one.

Sol

$$P_t = P_c \left(1 + \frac{m_a^2}{2} \right)$$
$$= 400 \left(1 + \frac{0.75^2}{2} \right) = 512.5 \text{ watts } \underline{\underline{\text{Ans}}}$$

Ex An AM broadcast radio transmitter radiates 10K watts of power if modulation percentage is 60. Calculate how much of this is the carrier power.

Sol

$$P_t = P_c \left(1 + \frac{m_a^2}{2} \right)$$
$$m_a = 60\% = \frac{60}{100} = 0.6, \quad P_t = 10 \text{ kWatts}$$
$$P_c = \frac{P_t}{1 + \frac{m_a^2}{2}} = \frac{10}{1 + \frac{0.6^2}{2}} = \frac{10}{1.18} = 8.471 \text{ kW } \underline{\underline{\text{Ans}}}$$

★ Current Calculation for Single-Tone AM

In AM, it is generally more convenient to measure the AM transmitter current than the power. In this case, the modulation index may be calculated from the values of unmodulated and modulated currents in the AM transmitter.

Mathematical Expression

Let I_c be the r.m.s value of unmodulated current (carrier) and I_t be the r.m.s value of modulated current (total) of an AM transmitter.

Let R be the antenna resistance through which these currents flow.

Power relation of single-tone modulation is given by,

$$P_t = P_c \left(1 + \frac{m_a^2}{2} \right) \quad \text{--- (1)}$$

where P_t = total or modulated power

P_c = carrier or unmodulated power

m_a = modulation index.

from eq (1)

$$\frac{P_t}{P_c} = 1 + \frac{m_a^2}{2}$$

$$\frac{I_t^2 \cdot R}{I_c^2 \cdot R} = 1 + \frac{m_a^2}{2}$$

$$\frac{I_t}{I_c} = \sqrt{1 + \frac{m_a^2}{2}}$$

$$I_t = I_c \sqrt{1 + \frac{m_a^2}{2}}$$

☆ Power content in Multiple-Tone Amplitude Modulation (AM)

A multiple-tone amplitude modulation is that type of modulation in which the modulating signal consists of more than one frequency components.

Mathematical Expression

→ Let us consider that a carrier signal $A \cos \omega_c t$ is modulated by a modulating signal $x(t)$, which is expressed as,

$$x(t) = V_1 \cos \omega_1 t + V_2 \cos \omega_2 t + V_3 \cos \omega_3 t \quad \text{--- (1)}$$

→ general expression for AM wave is

$$S(t) = A \cos \omega_c t + x(t) \cos \omega_c t \quad \text{--- (2)}$$

Put eq (1) in eq (2), we get.

$$S(t) = A \cos \omega_c t + [V_1 \cos \omega_1 t + V_2 \cos \omega_2 t + V_3 \cos \omega_3 t] \cos \omega_c t$$

$$S(t) = A \left[1 + \frac{V_1}{A} \cos \omega_1 t + \frac{V_2}{A} \cos \omega_2 t + \frac{V_3}{A} \cos \omega_3 t \right] \cos \omega_c t$$

we know that,

$$\frac{V}{A} = \frac{\text{max. amplitude of modulating sig}}{\text{Maximum amplitude of carrier sig}} = m_a, \text{ modulation index.} \quad \text{--- (3)}$$

$$S(t) = A [1 + m_1 \cos \omega_1 t + m_2 \cos \omega_2 t + m_3 \cos \omega_3 t] \cos \omega_c t \quad \text{--- (4)}$$

where $m_1 = \frac{V_1}{A}$, $m_2 = \frac{V_2}{A}$, $m_3 = \frac{V_3}{A}$ are modulation indexes of the corresponding frequency components.

$$S(t) = A \cos \omega_c t + m_1 A \cos \omega_c t \cos \omega_1 t + m_2 A \cos \omega_c t \cos \omega_2 t + m_3 A \cos \omega_c t \cos \omega_3 t$$

we know that total power in AM is given by,

$$P_t = P_c + P_s \quad \text{--- (A)}$$

→ carrier power P_c is given as.

$$P_c = (A \cos \omega_c t)^2 = \frac{A^2}{2}$$

Sideband power is given as,

$$P_s = \frac{1}{2} \overline{x^2(t)} = \frac{1}{2} \left[\overline{(V_1 \cos \omega_1 t)^2} + \overline{(V_2 \cos \omega_2 t)^2} + \overline{(V_3 \cos \omega_3 t)^2} \right] \quad \text{--- (5)}$$

But we know that,

$$m_1 = \frac{V_1}{A} \quad \text{so that } V_1 = m_1 A$$

$$m_2 = \frac{V_2}{A} \quad \text{so that } V_2 = m_2 A$$

$$m_3 = \frac{V_3}{A} \quad \text{so that } V_3 = m_3 A$$

→ Substituting the values of V_1, V_2 and V_3 in eq (5) we get

$$P_s = \frac{1}{2} \left[\overline{(m_1 A \cos \omega_1 t)^2} + \overline{(m_2 A \cos \omega_2 t)^2} + \overline{(m_3 A \cos \omega_3 t)^2} \right]$$

$$P_s = \frac{1}{2} \left[\frac{(m_1 A)^2}{2} + \frac{(m_2 A)^2}{2} + \frac{(m_3 A)^2}{2} \right]$$

$$= \frac{1}{4} A^2 [m_1^2 + m_2^2 + m_3^2]$$

→ Now putting the value of P_c and P_s in eq (A)

$$P_t = P_c + P_s$$

$$= \frac{A^2}{2} + \frac{1}{4} A^2 [m_1^2 + m_2^2 + m_3^2]$$

$$P_t = \frac{A^2}{2} \left[1 + \frac{1}{2} (m_1^2 + m_2^2 + m_3^2) \right]$$

$$= P_c \left[1 + \frac{m_1^2 + m_2^2 + m_3^2}{2} \right]$$

$$P_t = P_c \left[1 + \frac{m_1^2}{2} + \frac{m_2^2}{2} + \frac{m_3^2}{2} \right]$$

This expression may be extended upto to n -modulating term i.e.

$$P_t = P_c \left[1 + \frac{m_1^2}{2} + \frac{m_2^2}{2} + \frac{m_3^2}{2} + \dots + \frac{m_n^2}{2} \right]$$

Net modulation index

$$m_x = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots + m_n^2}$$

Ex An AM transmitter radiates 9 kW of power when the carrier is unmodulated and 10.125 kW when the carrier is sinusoidal modulated. Find the modulation index, percentage of modulation. Now, if another sine wave, corresponding to 40% modulation is transmitted simultaneously, then calculate the total radiated power.

Sol Single-tone sinusoidal amplitude modulation the total power is expressed as

$$P_t = P_c \left(1 + \frac{m_a^2}{2} \right)$$

$$P_t = 10.125 \text{ kW}, \quad P_c = 9 \text{ kW}$$

$$1 + \frac{m_a^2}{2} = \frac{P_t}{P_c}$$

$$\frac{m_a^2}{2} = 1.125 - 1$$

$$\frac{m_a^2}{2} = 0.125$$

$$m_a^2 = 2 \times 0.125$$

$$= 0.250$$

$$\boxed{m_a = 0.50} \quad \underline{\text{Am}}$$

$$m_x = \sqrt{m_1^2 + m_2^2}$$

$$m_1 = m_a = 0.5$$

$$m_2 = 0.4$$

$$m_x = \sqrt{(0.5)^2 + (0.4)^2}$$

$$= \sqrt{0.25 + 0.16} = \sqrt{0.41} = 0.64$$

$$P_t = P_c \left(1 + \frac{m_x^2}{2} \right) = 9 \times \left(1 + \frac{0.64^2}{2} \right)$$

$$= 9 (1 + 0.205)$$

$$\boxed{P_t = 10.84 \text{ kW}} \quad \underline{\text{Ans}}$$

★ Generation of AM Waves:-

→ The circuit that generates the AM waves is called as amplitude modulator.

→ There are two modulator circuits namely,

1) Square law modulator

2) Switching modulator.

→ Both modulators use non-linear element such as diode for their implementation. (A non-linear device is the device, with a non-linear relation b/w its current and voltage.)

→ Both modulators are low power modulator circuits.

→ AM modulators are non-linear devices with two inputs and one output.

→ One input is a single, high frequency carrier signal of constant amplitude and second input is comprised of relatively low-frequency information signals that may be a single frequency or a complex waveform made up of many frequencies.

1. Square Law Modulator

It consists of

- ① A non-linear device

- ② A band pass filter

- ③ A carrier source and modulating signal.

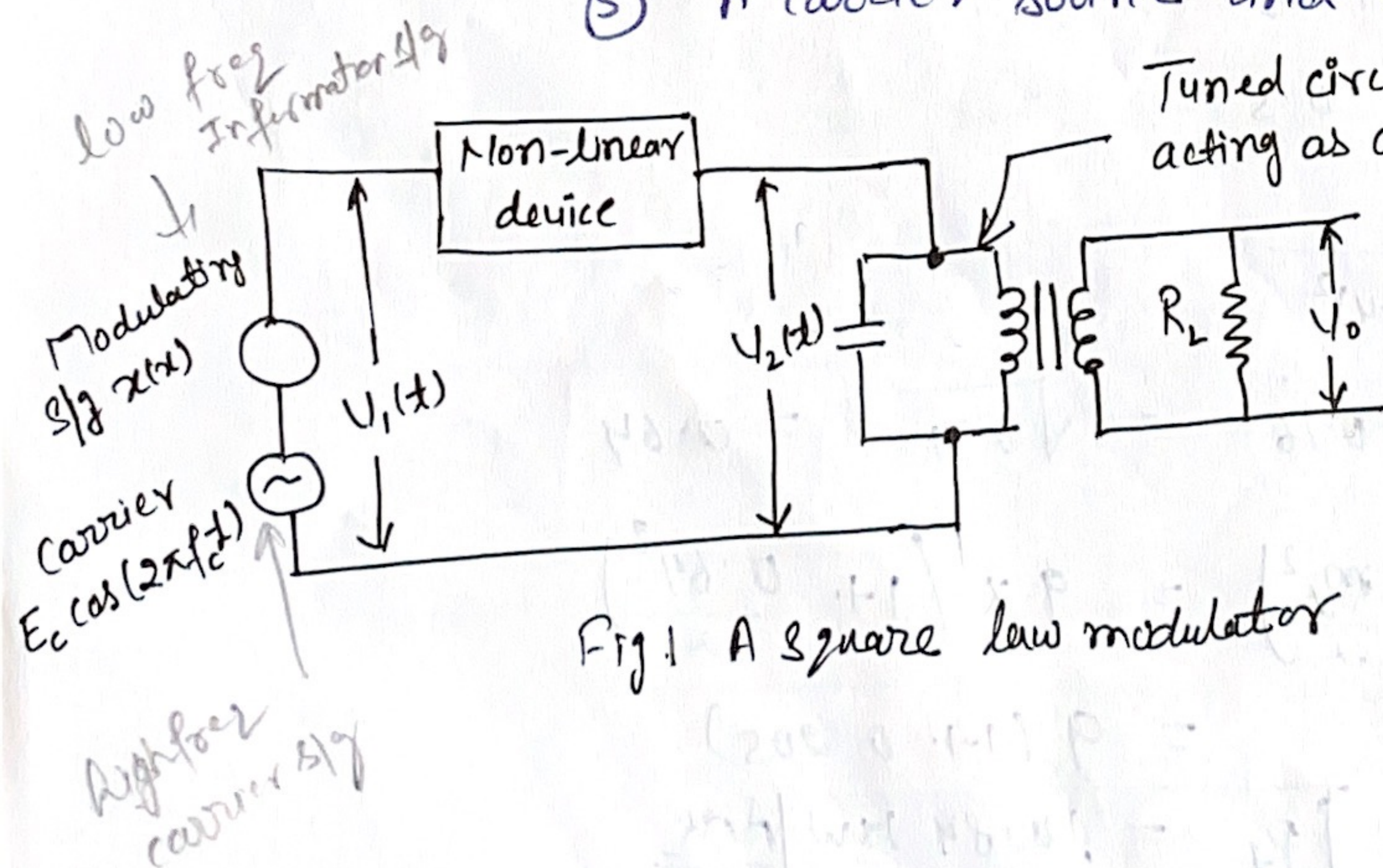
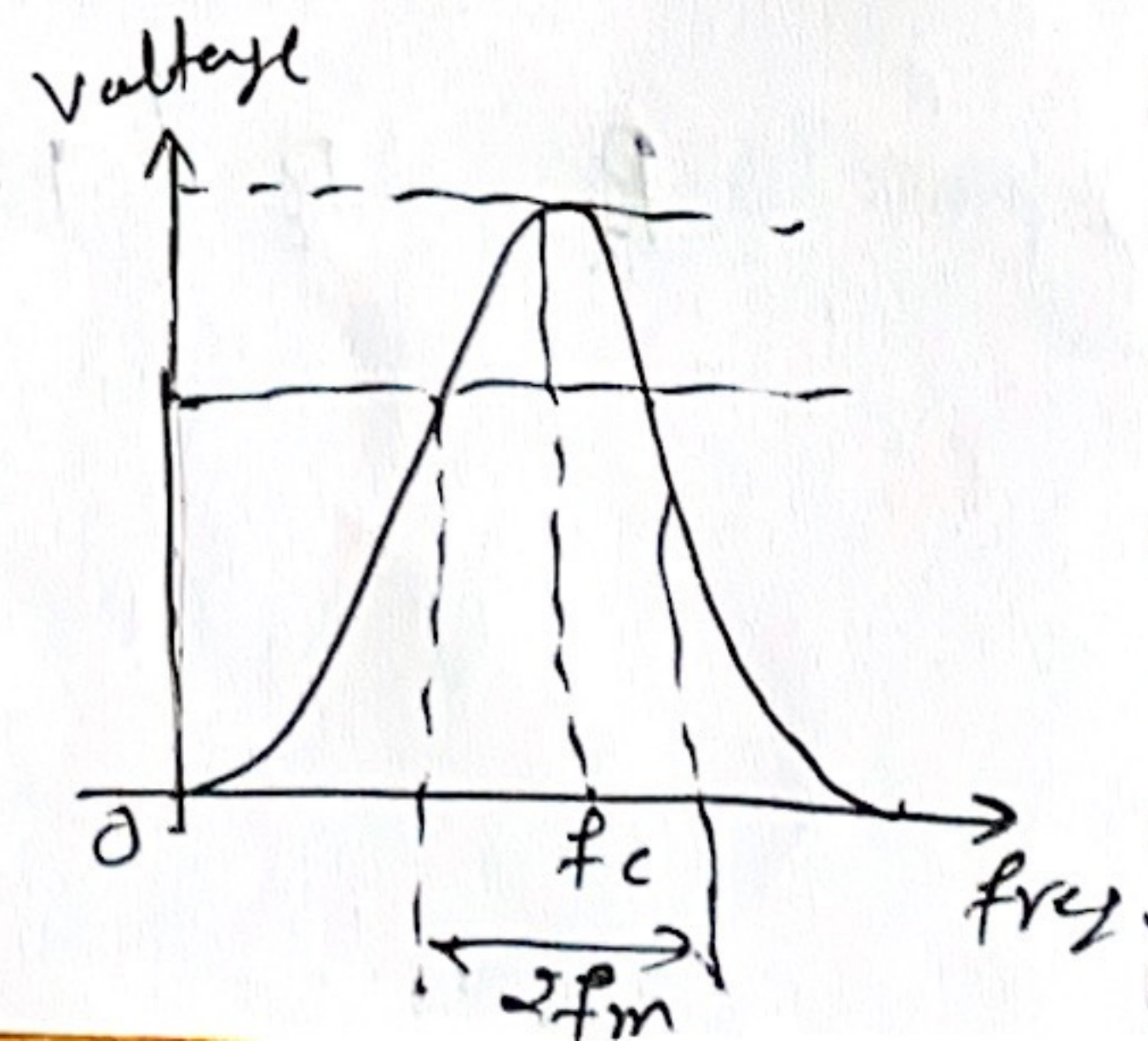


Fig. 1 A square law modulator



→ The modulating signal and carrier are connected in series with each other and their sum $V_1(t)$ is applied at the input of the non-linear device, such as diode, transistor etc.

$$V_1(t) = x(t) + E_c \cos(2\pi f_c t) \quad \text{--- (1)}$$

→ The input output relation for non-linear device is given by,

$$V_2(t) = aV_1(t) + bV_1^2(t) \quad \text{--- (2)}$$

where a and b are constants.

→ Substituting eq (1) in eq (2) we get,

$$V_2(t) = a[x(t) + E_c \cos(2\pi f_c t)] + b[x(t) + E_c \cos(2\pi f_c t)]^2$$

$$V_2(t) = ax(t) + aE_c \cos(2\pi f_c t) + b[x^2(t) + 2x(t)E_c \cos(2\pi f_c t) + E_c^2 \cos^2(2\pi f_c t)]$$

$$V_2(t) = \underbrace{ax(t)}_{(1)} + \underbrace{aE_c \cos(2\pi f_c t)}_{(2)} + \underbrace{bx^2(t)}_{(3)} + \underbrace{2bx(t)E_c \cos(2\pi f_c t)}_{(4)} + \underbrace{bE_c^2 \cos^2(2\pi f_c t)}_{(5)}$$

Term(1) $\Rightarrow ax(t) \rightarrow$ Modulating sig.

✓ Term(2) $\Rightarrow aE_c \cos(2\pi f_c t) \rightarrow$ carrier sig.

Term(3) $\Rightarrow bx^2(t) \rightarrow$ Squared modulating sig.

✓ Term(4) $\Rightarrow 2bx(t)E_c \cos(2\pi f_c t) \rightarrow$ AM wave with only sidebands

Term(5) $\Rightarrow bE_c^2 \cos^2(2\pi f_c t) \rightarrow$ Square carrier.

→ Out of these five terms, term 2 and 4 are useful whereas the remaining terms are not useful. Let us club terms 2, 4 and 1, 3, 5 as follow to get

$$\rightarrow V_2(t) = \underbrace{ax(t) + bx^2(t) + bE_c^2 \cos^2(2\pi f_c t)}_{\text{Unuseful Terms}} + \underbrace{aE_c \cos(2\pi f_c t) + 2bx(t)E_c \cos(2\pi f_c t)}_{\text{Useful Terms}}$$

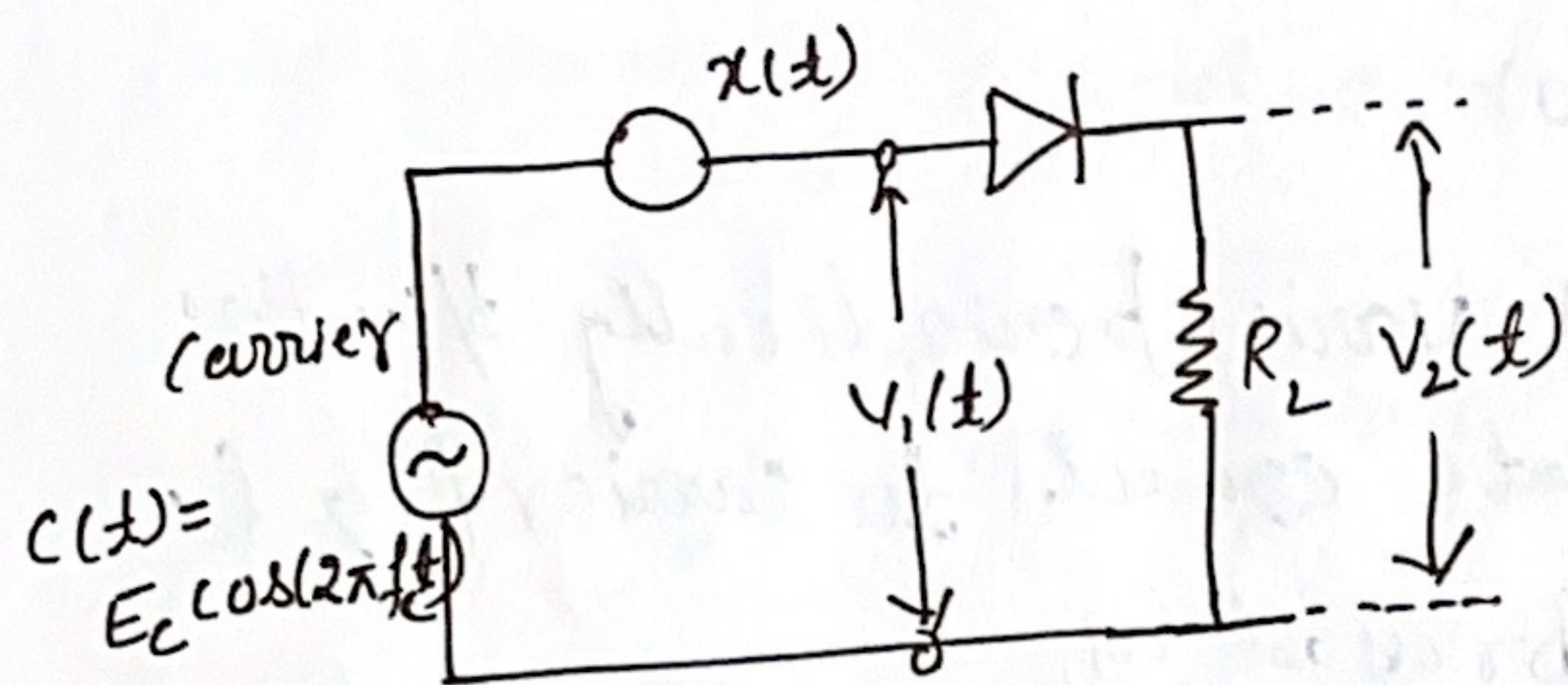
→ The LC tuned circuit acts as a bandpass filter. Its frequency response is shown in Fig 2. which shows that ckt is tuned to freq. f_c and its BW is equal to $2f_m$. This bandpass filter eliminates the unuseful terms from equation of $V_2(t)$. Hence o/p voltage $V_o(t)$ contains only useful terms. $V_o(t) = aE_c \cos(2\pi f_c t) + 2bx(t)E_c \cos(2\pi f_c t)$

$$\text{or } V_o(t) = [aE_c + 2bx(t)E_c] \cos(2\pi f_c t)$$

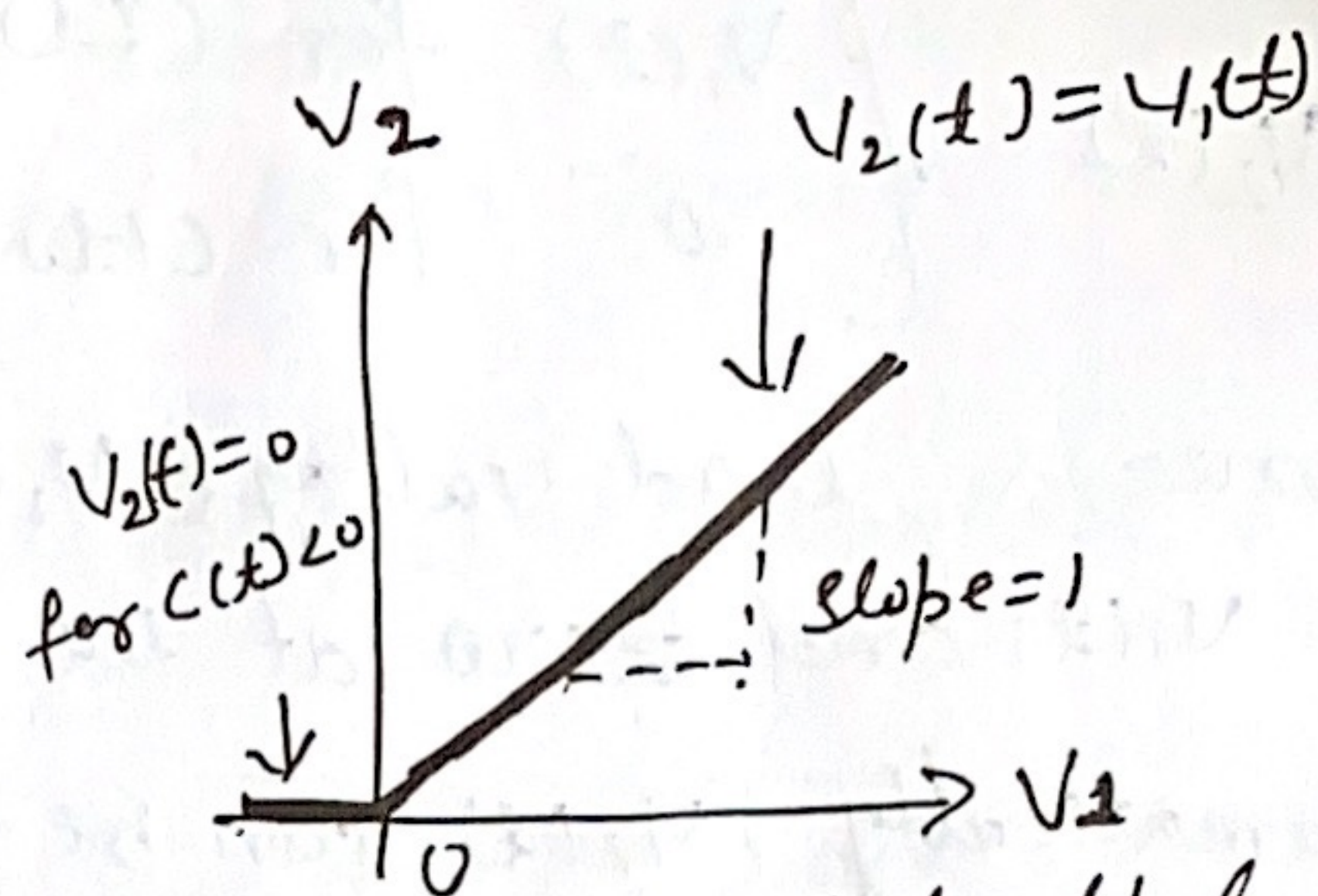
Therefore, $V_o(t) = aE_c \left[1 + \frac{2b}{a}x(t)\right] \cos(2\pi f_c t) \Rightarrow$ Comparing this expression with Standard AM. i.e.

$S(t) = E_c [1 + m x(t)] \cos(2\pi f_c t)$, we find that the expression for $V_o(t)$ of equation (3) represents an AM wave with $m = (2b/a)$. Hence, the square law modulator produces an AM wave.

2 Switching Modulator



(a) Switching modulator



(b) Idealized input output relation.

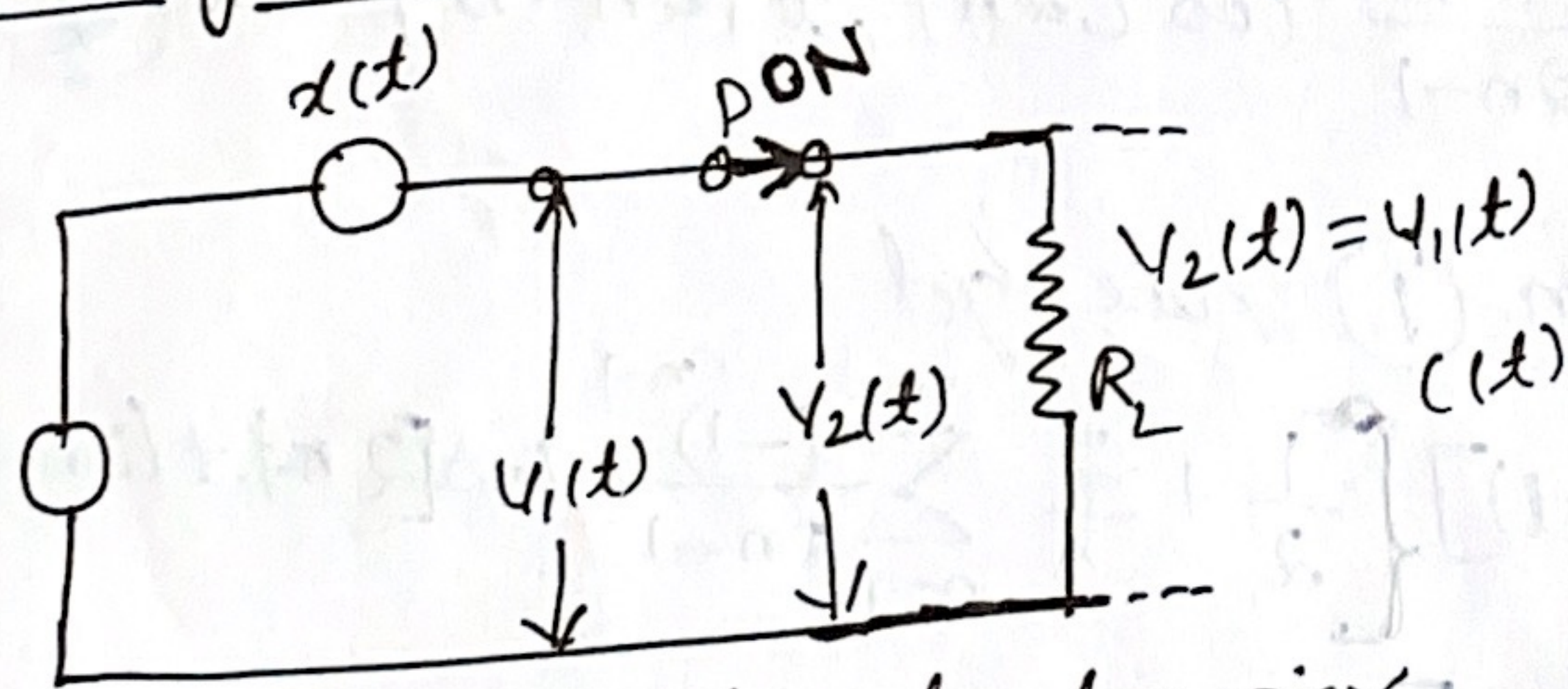
- The switching modulator using a diode has been shown in fig (a)
- This diode is assumed to be operating as a switch.
- The modulating signal $x(t)$ and the sinusoidal carrier signal $c(t)$ are connected in series with each other.

→ The input voltage to the diode is given by,

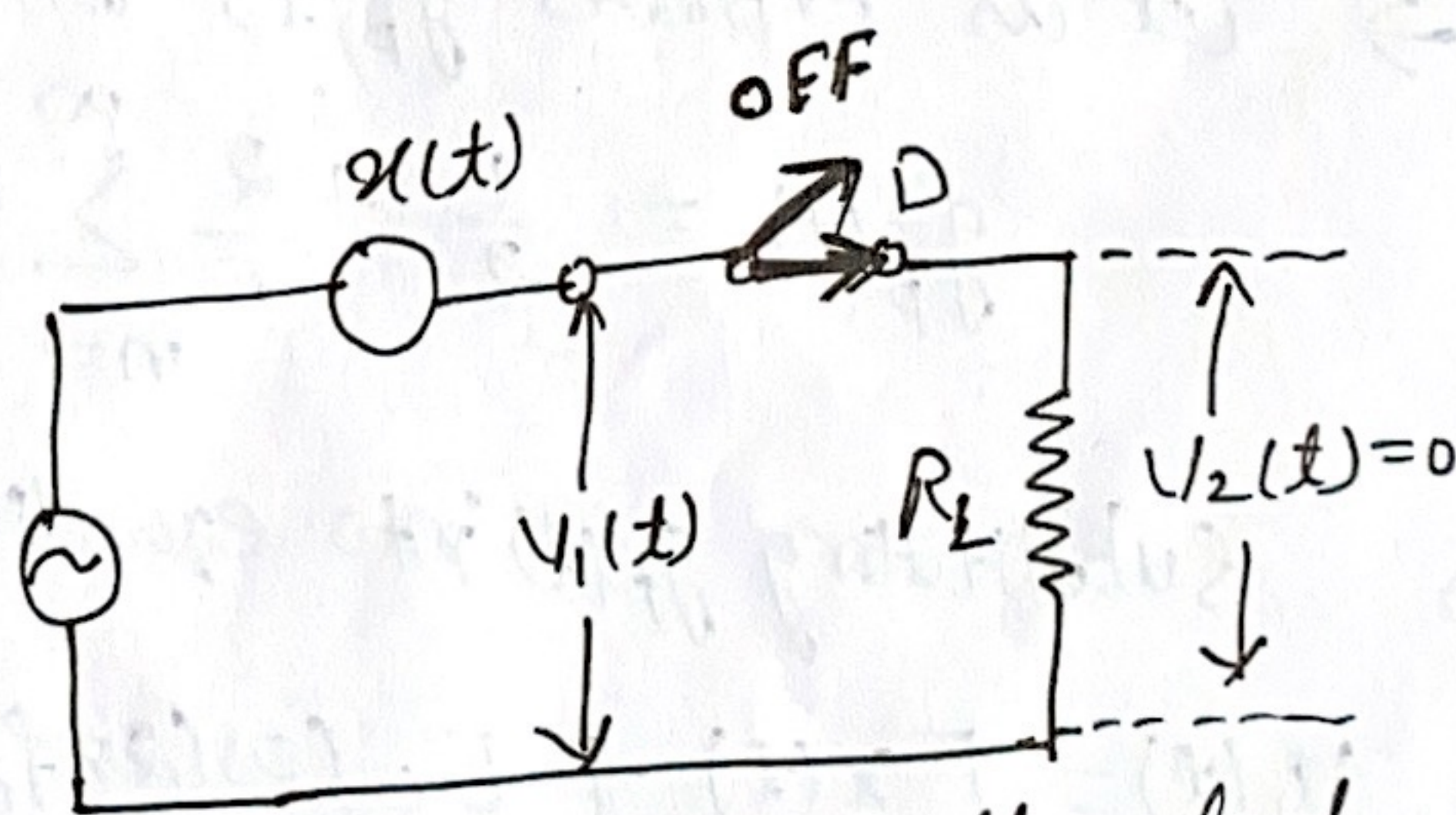
$$V_1(t) = c(t) + x(t) \\ = E_c \cos(2\pi f_c t) + x(t)$$

- The amplitude of carrier is much larger than that of $x(t)$
- $c(t)$ decides the status of the diode (ON or OFF)

Working Operation and Analysis :-



(a) In positive half cycle of carrier



(b) In negative half cycle of carrier

- Assume that diode acts as an ideal switch.
- In the pos half cycle of the carrier, diode is forward biased and it acts as a closed switch and offers zero impedance.
- In the -ve half cycle of the carrier, diode is reversed biased and it acts as a open switch and offers infinite impedance.
- Therefore, the o/p voltage $V_2(t) = V_1(t)$ in the (pos half cycle of $c(t)$)
- $V_2(t) = 0$ (-ve half cycle of $c(t)$)

Therefore,

$$V_2(t) = \begin{cases} V_1(t) & \text{for } c(t) > 0 \\ 0 & \text{for } c(t) < 0 \end{cases}$$

→ It means, load voltage $V_2(t)$ varies periodically w/ the value $V_1(t)$ and zero at the rate equal to carrier freq. f_c .

→ mathematically, $V_2(t)$ can be expressed as,

$$V_2(t) = V_1(t) \cdot g_p(t) \\ = [x(t) + E_c \cos(2\pi f_c t)] \cdot g_p(t) \quad \text{--- (1)}$$

where, $g_p(t)$ is a periodic pulse train of duty cycle to one half cycle period i.e. $T_0/2$ (where $T_0 = 1/f_c$)

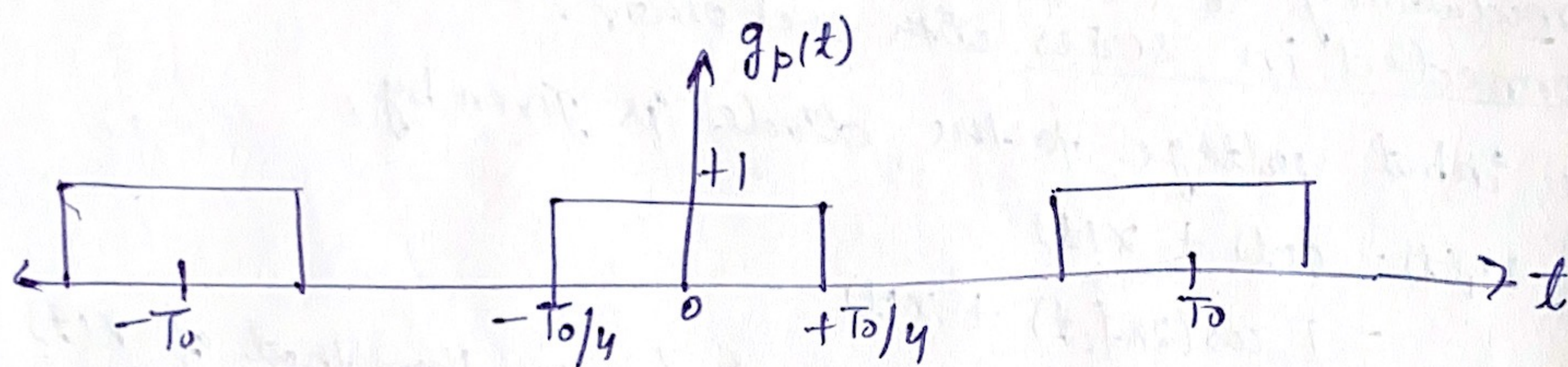


Fig. A periodic pulse train.

→ let us express $g_p(t)$ with the help of Fourier series as under.

$$g_p(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cos[2\pi f_c t (2n-1)] \quad \text{--- (2)}$$

→ Substituting $g_p(t)$ into equation (1), we get.

$$V_2(t) = [x(t) + E_c \cos(2\pi f_c t)] \left[\frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cos[2\pi f_c t (2n-1)] \right]$$

$$V_2(t) = [x(t) + E_c \cos(2\pi f_c t)] \left[\frac{1}{2} + \frac{2}{\pi} \cos(2\pi f_c t) + \text{odd harmonics} \right]$$

→ The odd harmonics in this expression are unwanted and therefore are assumed to be eliminated.

$$V_2(t) = \underbrace{\frac{1}{2} x(t)}_{\text{modulating sig}} + \underbrace{\frac{1}{2} E_c \cos(2\pi f_c t) + \frac{2}{\pi} x(t) \cos(2\pi f_c t)}_{\text{AM wave}} + \underbrace{\frac{2 E_c}{\pi} \cos^2(2\pi f_c t)}_{\text{second harmonic of carrier}}$$

→ In the expression, the first and fourth terms are unwanted terms whereas the second and third terms together represent AM wave.

$$V_z(t) = \frac{E_c}{2} \left[1 + \frac{\mu}{\pi E_c} \cos(2\pi f_c t) \right] \cos(2\pi f_c t) + \text{unwanted terms}$$

→ This is the required expression for the AM wave with $m = [\mu/\pi E_c]$

→ The unwanted terms can be eliminated using a B.P.F. (Band - Pass Filter).

☆ Detection (Demodulation) of AM waves:-

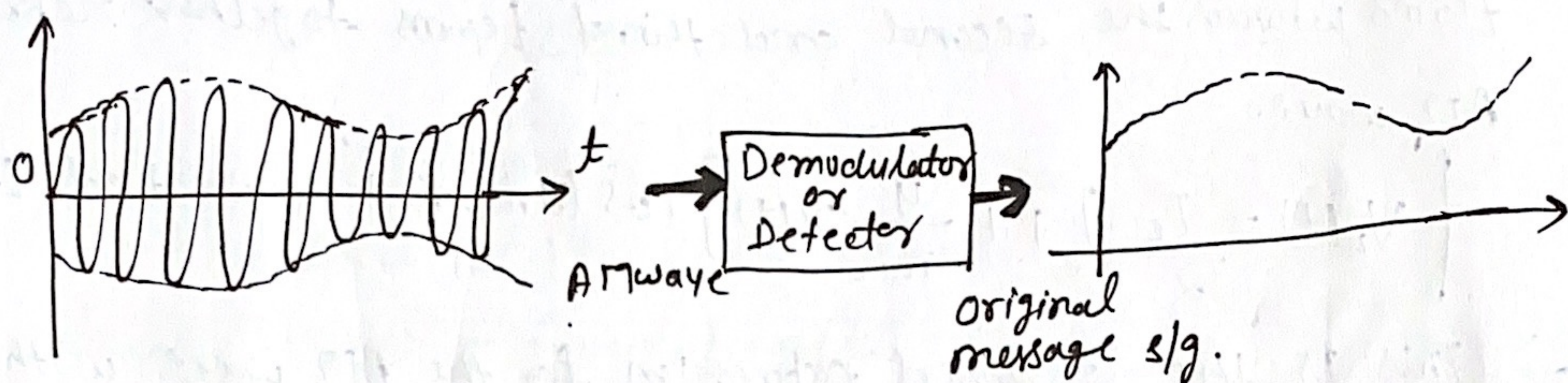


Fig. Process of demodulation or detection.

- AM demodulation is the reverse process of AM modulation.
- A conventional double-sideband AM receiver simply converts a received amplitude-modulated wave back to the original source information.
- To do this, a receiver must be capable of receiving, amplifying, and demodulating an AM wave.
- It must also be capable of bandlimiting the total radio-frequency spectrum to a specific desired band of frequencies. The selection process is called tuning the receiver.
- "The process of detection or demodulation is the process of recovering the message signal from the received modulated signal."

☆ Types of AM detectors:-

- (i) Square law detector
- (ii) Envelope detector.

1. Square Law Detector:-

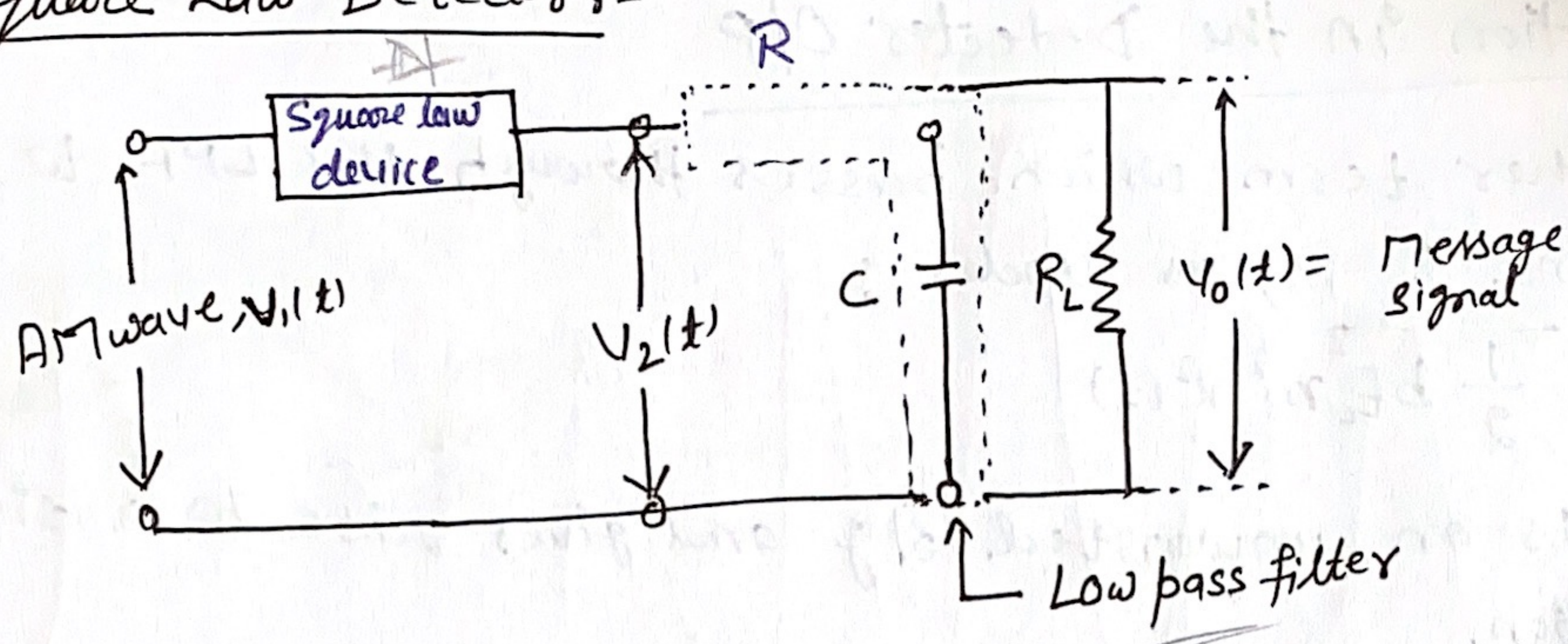


Fig. A square law detector

Working operation and Analysis :-

→ The input and output characteristics (+ transfer characteristics) of a square law device is non-linear and it is expressed mathematically as under;

$$V_2(t) = aV_1 + bV_1^2(t) \quad \text{--- (1)}$$

Where, $V_1(t)$ = input voltage to the detector = AM wave

$$\text{Therefore, } V_1(t) = E_c [1 + m x(t)] \cos(2\pi f_c t) \quad \text{--- (2)}$$

Substituting for $V_1(t)$ in equation (1)

$$V_2(t) = a E_c [1 + m x(t)] \cos(2\pi f_c t) + b E_c^2 [1 + m x(t)]^2 \cos^2(2\pi f_c t) \quad \text{--- (3)}$$

{ We know that, $\cos^2 \theta = \frac{1}{2} [1 + \cos 2\theta]$

$$\text{Therefore, } \cos^2(2\pi f_c t) = \frac{1}{2} [1 + \cos(4\pi f_c t)] \quad \text{--- (4)}$$

Putting (4) in eq (3), we get

$$V_2(t) = a E_c [1 + m x(t)] \cos(2\pi f_c t) + \frac{b E_c^2}{2} [1 + 2m x(t) + m^2 x^2(t)] [1 + \cos(4\pi f_c t)]$$

→ Out of these terms, the only desired term is $b E_c^2 m x(t)$ which is due to the bV_1^2 term. Hence, the name of this detector is square law detector. --- (5)

→ This desired term is extracted by using a Low pass filter (LPF) after the diode. Thus, after the LPF, we get

$$V_0(t) = (b E_c^2 m) x(t) \quad \text{--- (6)}$$

→ This means that we have recovered the message s/g $x(t)$ at the output of the detector.

Distortion in the Detector O/P

→ An other term which passes through the LPF to Load Resistance R_L is,

$$\frac{1}{2} b E_c^2 m^2 x^2(t)$$

→ This is an unwanted s/g and gives rise to a signal distortion.

$$\rightarrow \text{Ratio} = \frac{\text{Desired o/p}}{\text{Undesired o/p}} = \frac{b E_c^2 m x(t)}{\frac{1}{2} b E_c^2 m^2 x^2(t)} = \frac{2}{m x(t)}$$

→ We should maximize this ratio in order to minimize the distortion.

→ To achieve this, we should choose $|m x(t)|$ small as compared to unity(1) for all values of t . If 'm' is small, then, the AM wave is weak.

2 Envelope Detector

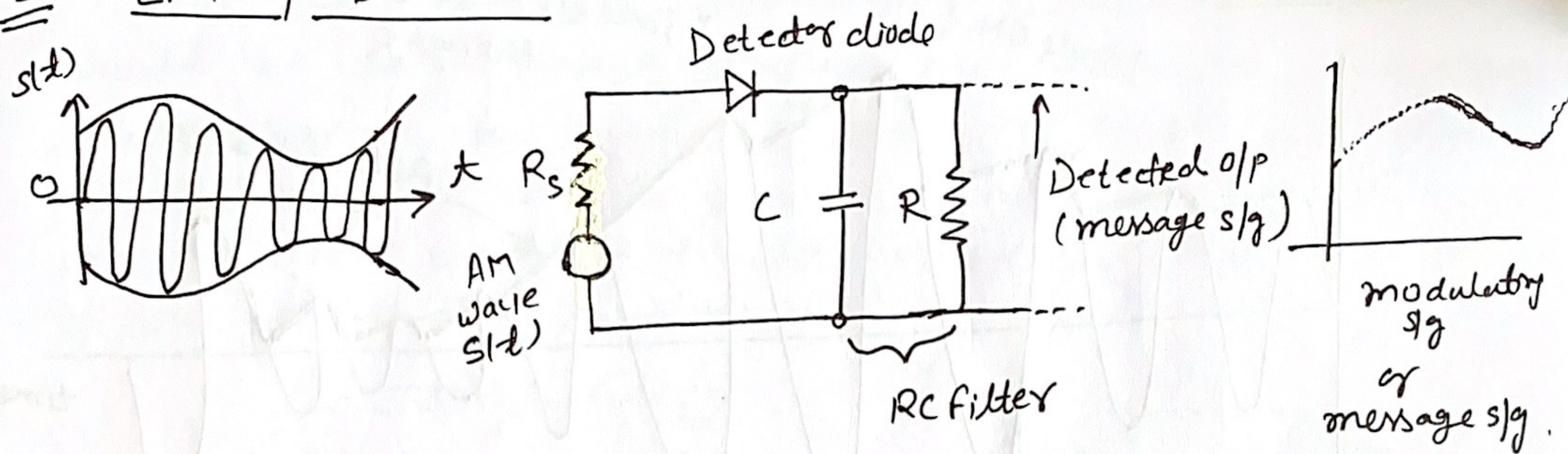


Fig 1. Envelope detector for detection of AM wave.

- The envelope detector is a simple and very efficient device which is suitable for the detection of a narrowband AM s/g.
- A narrowband AM wave is the one in which, the carrier freq. f_c is much higher as compared to the B.W of the modulating s/g.
- It produces an o/p s/g that follows the envelope of the i/p AM signal exactly.
- It is used in all the commercial AM radio receivers.

Working Operation :-

- The amplitude modulated wave (AM) is applied at the i/p of the detector.
- In every +ve half cycle of the i/p, the detector diode is forward biased.
- It will charge the filter capacitor C connected across the load resistance R to almost the peak value of the i/p voltage.
- As soon as the capacitor charges to the peak value, the diode stops conducting.
- The capacitor will discharge through R b/w the +ve peaks as shown in Fig 2.
- The discharging process continues until the next +ve half cycle.
- When the i/p s/g becomes greater than the capacitor voltage, the diode conducts again and the process repeats itself.

Waveforms:-

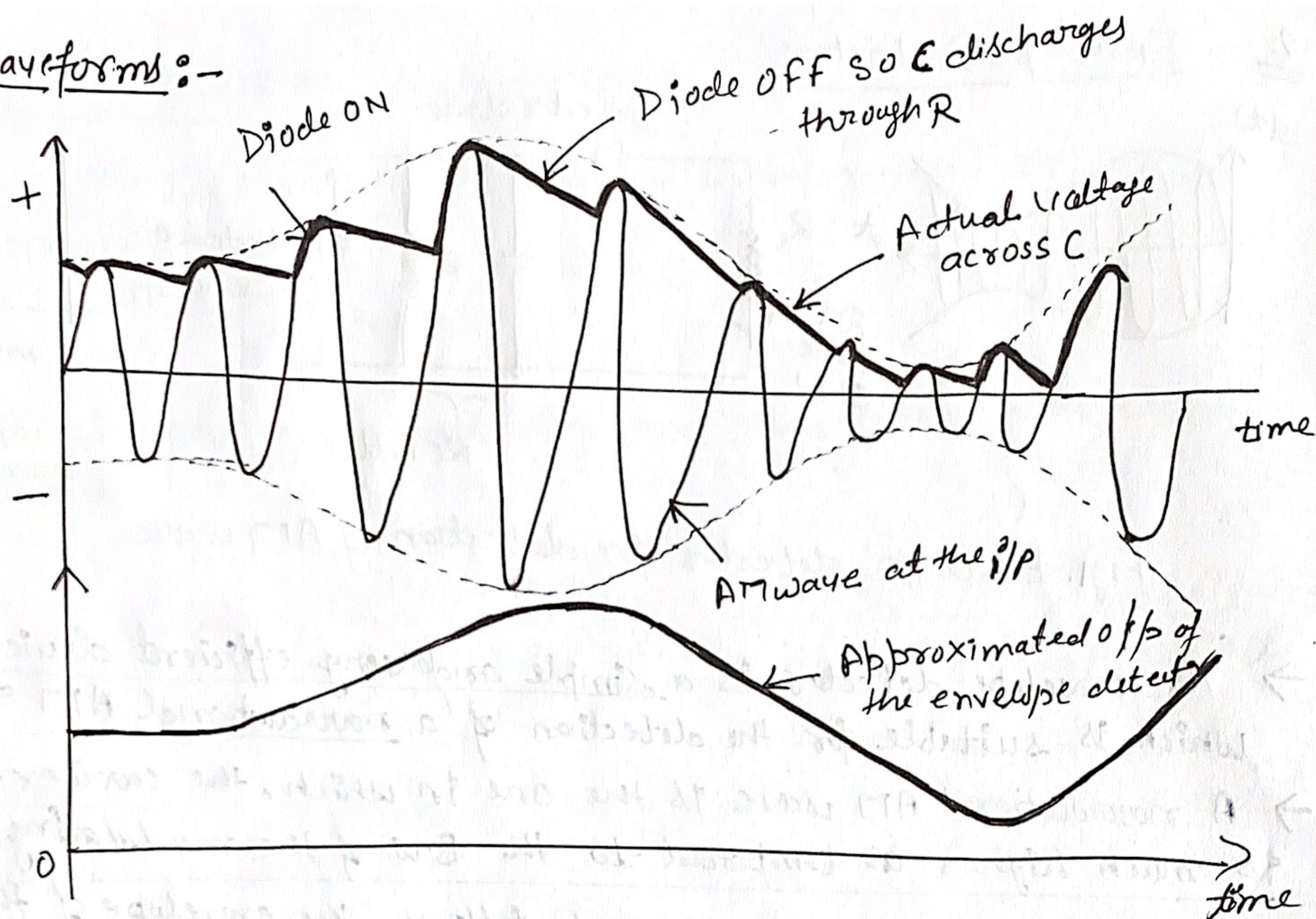


Fig 2 Input-Output waveforms for an envelope detector

- The i/p - output waveforms for the envelope detector are shown in Fig 2.
- It shows charging discharging of the filter capacitor and the approximated o/p voltage.
- We assume that the diode is ideal which presents a zero resistance when it is ON and infinite resistance when it is OFF. Also assumed that the AM wave applied to the i/p of the detector is supplied by a source having internal resistance R_s .

Selection of the RC time constants

- The capacitor charges through D and R_s when the diode is ON and it discharges through R when the diode is OFF. The charging time constant $R_s C$ should be short compared to the carrier period $1/f_c$.

Thus, $R_s C \ll \frac{1}{f_c}$

→ On the other hand, the discharging time constant RC should be long enough so that the capacitor discharges slowly through the load resistance R . But this time constant should not be too long which will not allow the capacitor voltage to discharge at the max. rate of change of the envelope.

$$\frac{1}{f_c} \ll RC \ll \frac{1}{W}$$

Where, $W = \text{max. modulating freq.}$

Distortions in Envelope detector O/P

① Diagonal clipping:-

This type of distortion occurs when the RC time const. of the load ckt is too long. Due to this, the RC circuit cannot follow the fast changes in the modulating envelope.

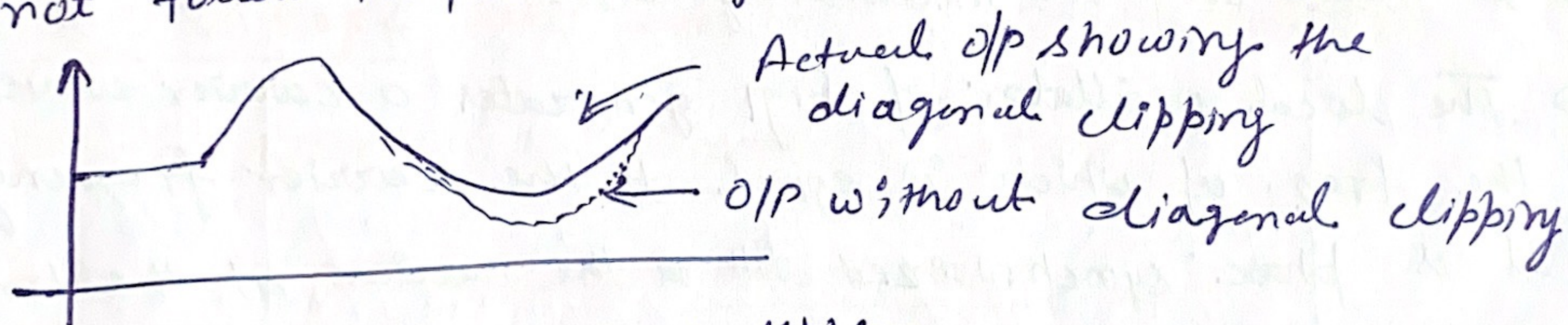
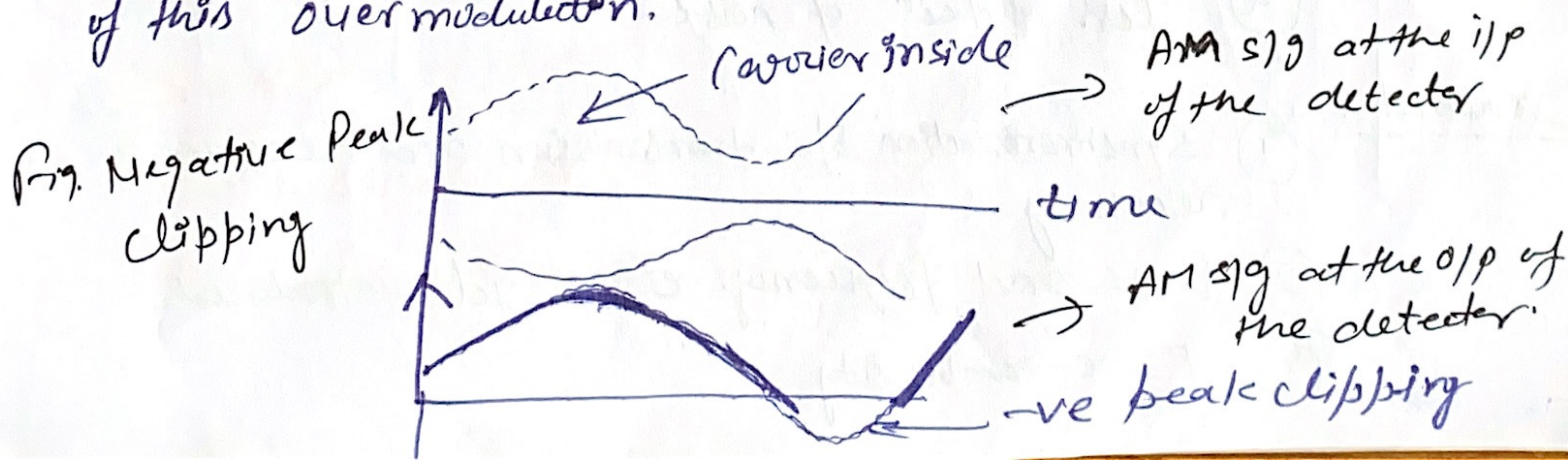


Fig. Diagonal clipping.

② Negative peak clipping

This distortion occurs due to the fact that the modulation index on the o/p side of the detector is higher than that on its i/p side. Hence, at higher depths of modulation of the transmitted sig, the overmodulation may take place at the o/p of the detector.

→ The -ve peak clipping will take place as a result of this overmodulation.



3 Synchronous Detection :-

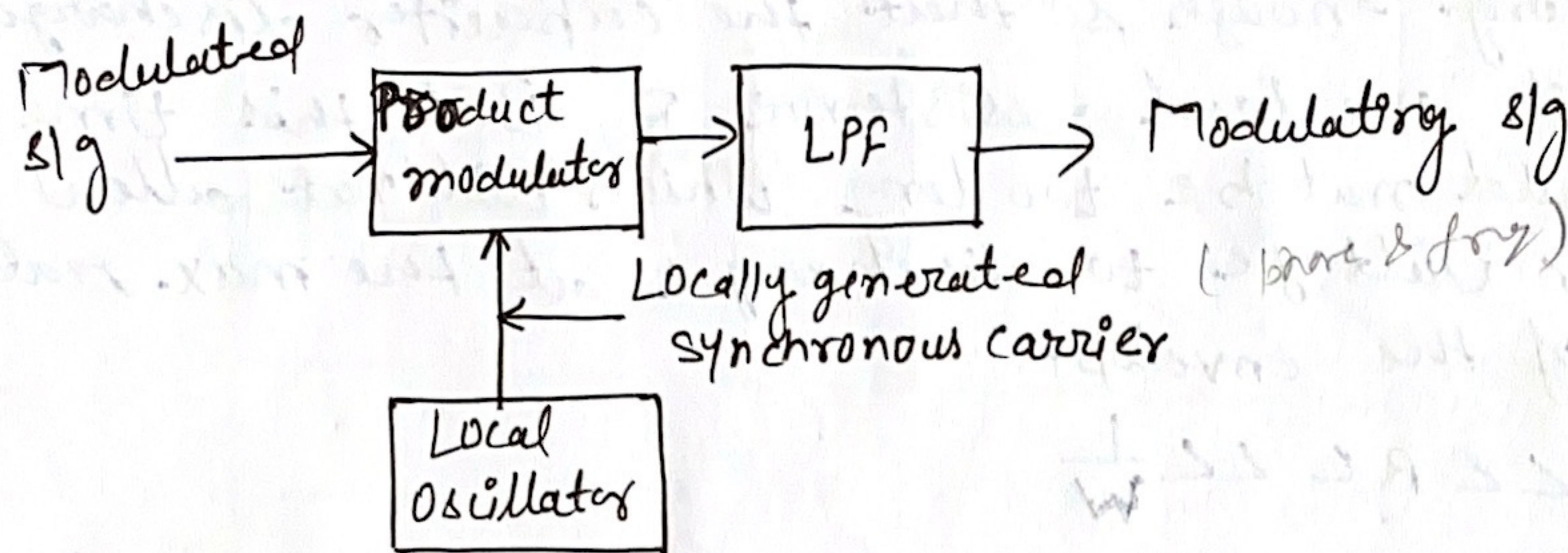


Fig.1 Synchronous Detection

- The principle of synchronous detection states that the i/p modulated sig is multiplied with a locally generated, phase and frequency, synchronous carrier at the receiver.
- The product is passed through a L.PF (Low Pass Filter) to pass only the modulating signals.
- The local oscillator of Fig.1 generates a carrier wave, the freq. of which is equal to the carrier frequency and it is phase synchronized with the carrier at the transmitter.
- Product Modulator is basically a multiplier which multiplies the i/p modulated sig and the locally generated carrier.
- At the output of the multiplier we obtain various freq. components one of which is the modulating (baseband) sig. It is allowed to pass through by the LPF.

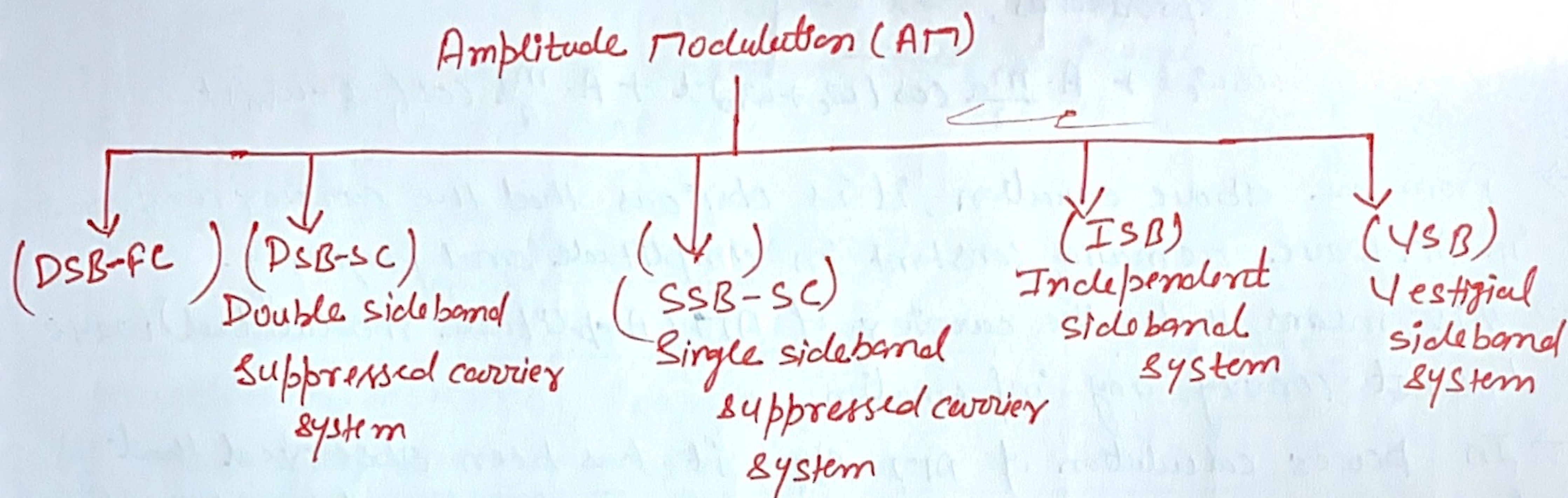
Advantages

- ① Better quality of demodulation
- ② Less effect of noise.

Drawbacks

- ① Synchronization b/w transmission and receiver is necessary.
- ② Phase and frequency errors get introduced.
- ③ More complexity.

★ Other Types of AM



★ Disadvantages of DSB-FC Transmission

- ① Power wastage takes place in DSB-FC transmission.
- ② DSB-FC system is B.W inefficient system
- ③ AM wave gets affected due to noise

★ Features of AM:-

- ① AM transmitters are less complex
- ② AM receivers are simple, detection is easy.
- ③ AM receivers are cost efficient. Hence, even a common person can afford to buy it
- ④ AM wave can travel a longer distance.
- ⑤ Low B.W.

★ Applications of AM

- ① Radio broadcasting
- ② Picture transmission in a TV system